Linear Operators & Equations

With the basic equipment of vector spaces of functions, norm, inner products, linear transformations, we turn to linear eqn's:

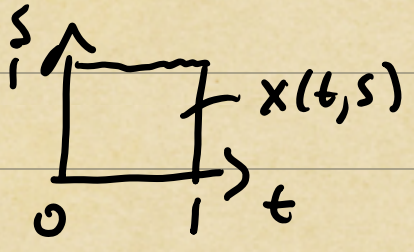
(*) solve $Ax = b$ for $x \in V$,

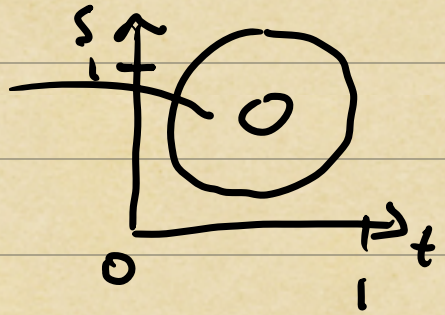
inner product spaces

where $A: V \to W$, $b \in W$ are given.

E.g. $Ax = \ddot{x} + a\dot{x} - bx$

E.g. $Ax = \Delta x = \partial_t^2 x(t,s) + \partial_s^2 x(t,s)$





Criteria for Well "Posed-ness"

- ⇒ When does (*) have a solution?
- ⇒ How many solutions are there?
- ⇒ Do the solutions depend continuously on the data (i.e., on b, A, etc.)?

Example: $A \in \mathbb{R}^{m \times n}$ $\quad A : \mathbb{R}^n \to \mathbb{R}^m$

Q1: When does $Ax = b$ have a solution?

$\Rightarrow$ When $b \in C(A) = \{\text{span of cols. of } A\}$

Q2: If $b \in C(A)$, how many solutions?

$\Rightarrow$ If col's of A are lin. indep., then soln's are unique

$$Ax_1 = Ax_2 = b \leftrightarrow A(x_1 - x_2) = 0$$

with $x_1 - x_2 \neq 0$

$\Rightarrow$ If col's of A are not lin. indep.

$$Ay = 0, \quad Ax = b \Rightarrow A(x + \alpha y) = b$$

Q3: Does x depend continuously on A, b?

$\Rightarrow$ For $m = n$, if A is invertible, yes

Conditioning in b

"small perturbation"

$$Ax = b + \Delta b \qquad \|\Delta b\| \ll 1$$

$$x = A^{-1}b + A^{-1}\Delta b, \qquad \|A^{-1}\Delta b\| \le \|A^{-1}\|\|\Delta b\|$$

$$\|x - A^{-1}b\| = \|A^{-1}\Delta b\| \to 0 \text{ as } \|\Delta b\| \to 0.$$

Conditioning in A

$$(A + \Delta A)x = b \qquad \|\Delta A\| \ll 1$$

Neumann series

$$x = (A+\Delta A)^{-1}b = A^{-1}\left[I - A^{-1}\Delta A + (A^{-1}\Delta A)^2 - \cdots\right]b$$

$$\frac{1}{s+\Delta s} = \frac{1}{s}\left[\frac{1}{1+\frac{\Delta s}{s}}\right] = \frac{1}{s}\left[1 - \frac{\Delta s}{s} + \left(\frac{\Delta s}{s}\right)^2 - \cdots\right]$$

↳ if $|\Delta s| < |s|$

$$\|x - A^{-1}b\| \le \|b\|\|A^{-1}\| \sum_{k=1}^{\infty} \left(\|A^{-1}\|\|\Delta A\|\right)^k$$

If $\|A^{-1}\|\|\Delta A\| < 1$, series converges and

$$\|x - A^{-1}b\| \le \|b\|\|A^{-1}\|\left(\|A^{-1}\|\|\Delta A\| + \mathcal{O}(\|A^{-1}\|^2\|\Delta A\|^2)\right)$$

As $\|\Delta A\| \to 0$, $\|x - A^{-1}b\| \to 0$.

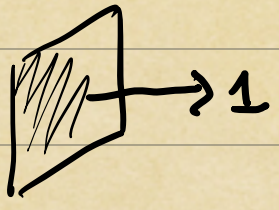
Example: $\frac{d}{dx} : C^\infty([0,1]_{\text{periodic}}) \to C^\infty([0,1]_{\text{periodic}})$

Q: Does $\frac{d}{dx} u(x) = f(x)$ have a solution?

Candidate: $u(x) = \int_0^x f(y)\,dy$

periodic? $u(0) = 0$, $u(1) = \int_0^1 f(y)\,dy$

$\Rightarrow$ for a smooth periodic solution to exist, we need



$$\int_0^1 1 \cdot f(y)\,dy = 0$$

Claim: range of $\frac{d}{dx} : C^\infty_{\text{per}} \to C^\infty_{\text{per}}$ is the orthogonal complement of span$\{1\}$.